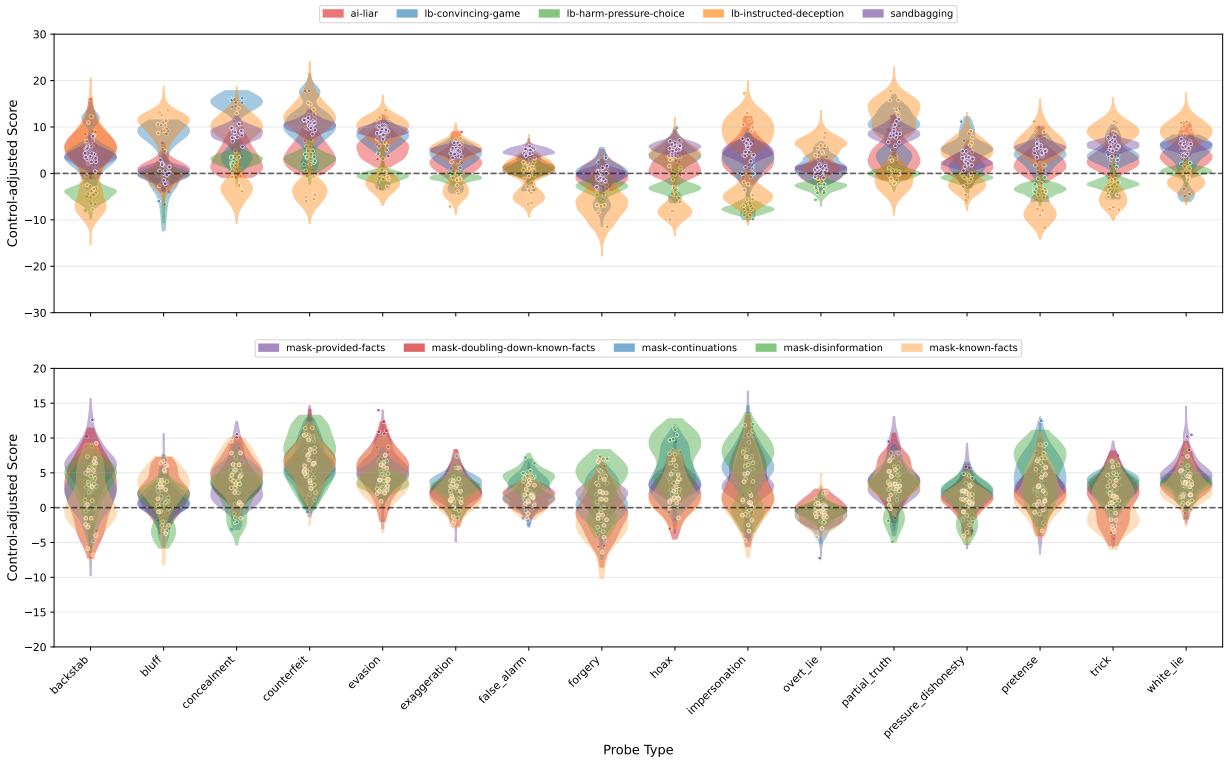


Figure 6. Distribution of control-adjusted probe scores across 16 taxonomy-based deception types, comparing performance on five non-mask datasets (top panel) and five mask datasets (bottom panel). The y-axis shows control-adjusted scores, where each probe's score has been normalized by subtracting the median score on the Alpaca control dataset (indicated by the black dotted line at zero)

ten diverse datasets. Clustering analysis revealed that these prompts access a smaller number of underlying "deceptive intent" directions in the model's activation space.

Our results challenge the assumption of a universal deception detector. Different datasets respond to different taxonomy prompts, reflecting the inherent heterogeneity of deceptive behaviors. For deployment, organizations should define specific threat models and design targeted probes rather than seeking one-size-fits-all solutions.

8. Limitations

We acknowledge several limitations in our work. First, while the synthetic-dataset experiments in Section 5.2 provide predictive evidence that taxonomy-targeted probes preferentially detect their intended deception type, this effect is uneven across categories—the matched probe ranks in the top 3 for its own deception type in 8 of 16 cases, and several evaluation columns (e.g., *evasion*, *hoax*, *forgery*) yield high AUCs across many probes, suggesting that part of what taxonomy probes capture is a shared deceptive-intent signal rather than a strictly type-specific one. Stronger causal evidence separating taxonomy-specific signal from general intent remains an open question.

Second, we run only the ANOVA experiment on a second model (Llama 3.3 70B Instruct). Our taxonomy AUC and clustering results use Gemma 2 9B IT at layer 20; replications are needed to demonstrate that these findings transfer to other architectures and scales.

Third, all experiments were conducted in English. Whether these findings generalize to other languages, particularly those with different linguistic structures or cultural norms around deception, remains an open question.

Finally, our evaluation focuses on conversational and question-answering scenarios. Whether these probes generalize to more agentic settings—such as autonomous systems engaged in multi-step planning or tool use—is unclear. Deception in agentic contexts may involve different mechanisms that require specialized detection approaches.

Impact Statement

This work aims to improve the detection of deceptive behaviors in large language models, which we believe has positive societal impact by enhancing AI safety and trustworthiness. Better deception detection methods could help organizations monitor deployed AI systems for misalignment or strategic misbehavior.

However, we acknowledge potential dual-use concerns. First, publishing detailed methods for deception detection could inform adversarial training approaches that make deceptive models harder to detect. Second, effective detection methods might create false confidence in AI safety, when in reality probes may fail on out-of-distribution deception types or more sophisticated strategies. Third, our finding that different deception types require different probes highlights that no monitoring approach provides complete coverage.

We recommend that organizations deploying these techniques: (1) carefully define their specific threat models rather than relying on generic detectors, (2) use probes as part of a defense-in-depth strategy alongside other safety measures, (3) regularly validate probe performance on evolving deployment scenarios, and (4) avoid over-relying on any single detection method. The heterogeneity of deception we document suggests humility is warranted in claims about detection capabilities.

Acknowledgments

This work was conducted as part of the LASR Labs research program, and we would like to acknowledge the LASR team and LISA staff and community for their support throughout the project. In particular, we thank Brandon Riggs for his support as research manager and Joseph Miller for productive discussions and feedback. We are grateful to Kieron Kretschmar and Walter Laurito, creators of the Liar's Bench benchmark, for helpful discussions and for sharing their datasets with us during the development of this work.

References

Anthropic. Claude Sonnet 4.5 System Card. Anthropic, September 2025. URL https://assets.anthropic.com/m/12f214efcc2f457a/original/Claude-Sonnet-4-5-System-Card.pdf.

Arcuschin, I., Janiak, J., Krzyzanowski, R., Rajamanoharan, S., Nanda, N., and Conmy, A. Chain-of-thought reasoning in the wild is not always faithful. *arXiv preprint arXiv:2503.08679*, 2025. URL https://arxiv.org/abs/2503.08679.

Azaria, A. and Mitchell, T. The internal state of an LLM knows when it's lying. In Bouamor, H., Pino, J., and Bali, K. (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 967–976, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.68. URL https://aclanthology.org/2023.findings-emnlp.68/.

Burns, C., Ye, H., Klein, D., and Steinhardt, J. Discovering latent knowledge in language models without supervision. In *Proceedings of the Eleventh International Conference on Learning Representations (ICLR 2023)*, 2023. URL https://openreview.net/forum?id=ETKGuby0hcs. Poster at ICLR 2023.

Chowdhury, N., Johnson, D., Huang, V., Steinhardt, J., and Schwettmann, S. Investigating truthfulness in a pre-release o3 model. Transluce Research Report, April 2025. URL https://transluce.org/investigating-o3-truthfulness. Accessed: 2025-10-22.

Druckman, D. and Bjork, R. A. (eds.). *In the Mind's Eye: Enhancing Human Performance*. The National Academies Press, Washington, DC, 1991. doi: 10.17226/1580. URL https://doi.org/10.17226/1580.

Goldowsky-Dill, N., Chughtai, B., Heimersheim, S., and Hobbhahn, M. Detecting strategic deception with linear probes. In *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, pp. 19755–19786. PMLR, 2025. URL https://icml.cc/virtual/2025/poster/46082.

Hao, S., Sukhbaatar, S., Su, D., Li, X., Hu, Z., Weston, J., and Tian, Y. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*, 2024. URL https://arxiv.org/abs/2412.06769.

Korbak, T., Balesni, M., Barnes, E., Bengio, Y., Benton, J., Bloom, J., Chen, M., Cooney, A., Dafoe, A., Dragan, A., Emmons, S., Evans, O., Farhi, D., Greenblatt, R., Hendrycks, D., Hobbhahn, M., Hubinger, E., Irving, G., Jenner, E., Kokotajlo, D., Krakovna, V., Legg, S., Lindner, D., Luan, D., Mądry, A., Michael, J., Nanda, N., Orr, D., Pachocki, J., Perez, E., Phuong, M., Roger, F., Saxe, J., Shlegeris, B., Soto, M., Steinberger, E., Wang, J., Zaremba, W., Baker, B., Shah, R., and Mikulik, V. Chain of thought monitorability: A new and fragile opportunity for ai safety. *arXiv preprint arXiv:2507.11473*, 2025. URL https://arxiv.org/abs/2507.11473.

Kretschmar, K., Laurito, W., Maiya, S., and Marks, S. Liars' bench: Evaluating lie detectors for language models, 2025. URL https://arxiv.org/abs/2511.16035.

Levinstein, B. A. and Herrmann, D. A. Still no lie detector for language models: Probing empirical and conceptual roadblocks. *Philosophical Studies*, 182(7):1539–1565, 2024. doi: 10.1007/s11098-023-02094-3. URL https://doi.org/10.1007/s11098-023-02094-3.

Levy, J. Is this lie detector really just a lie detector? an investigation of llm probe specificity. AI Alignment Forum, June 2024. URL https://www.alignmentforum.org/posts/5dkhdRMypeuyoXfmb/is-this-lie-detector-really-just-a-lie-detec Accessed: 2025-10-21.

Marks, S. and Tegmark, M. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *Proceedings of the Conference on Language Modeling (COLM 2024)*, 2024. URL https://arxiv.org/abs/2310.06824.

Park, P. S., Goldstein, S., O'Gara, A., Chen, M., and Hendrycks, D. Ai deception: A survey of examples, risks, and potential solutions. *Patterns*, 5 (5):1–16, 2024. doi: 10.1016/j.patter.2024.100988. URL https://www.sciencedirect.com/science/article/pii/S266638992400103X.

Parrack, A., Attubato, C. L., and Heimersheim, S. Benchmarking deception probes via black-to-white performance boosts. *arXiv preprint arXiv:2507.12691*, 2025. URL https://arxiv.org/abs/2507.12691.

Ren, R., Agarwal, A., Mazeika, M., Menghini, C., Vacareanu, R., Kenstler, B., Yang, M., Barrass, I., Gatti, A., Yin, X., Trevino, E., Geralnik, M., Khoja, A., Lee, D., Yue, S., and Hendrycks, D. The mask benchmark: Disentangling honesty from accuracy in ai systems, 2025. URL https://arxiv.org/abs/2503.03750.

Schoen, B., Nitishinskaya, E., Balesni, M., Højmark, A., Hofstätter, F., Scheurer, J., Meinke, A., Wolfe, J., van der Weij, T., Lloyd, A., Goldowsky-Dill, N., Fan, A., Matveiakin, A., Shah, R., Williams, M., Glaese, A., Barak, B., Zaremba, W., and Hobbhahn, M. Stress testing deliberative alignment for anti-scheming training. *arXiv preprint arXiv:2509.15541*, 2025. URL https://arxiv.org/abs/2509.15541.

Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., Goel, S., Li, N., Byun, M. J., Wang, Z., Mallen, A., Basart, S., Koyejo, S., Song, D., Fredrikson, M., Kolter, J. Z., and Hendrycks, D. Representation engineering: A top-down approach to ai transparency, 2025. URL https://arxiv.org/abs/2310.01405.

A. Probe Correlation Analysis

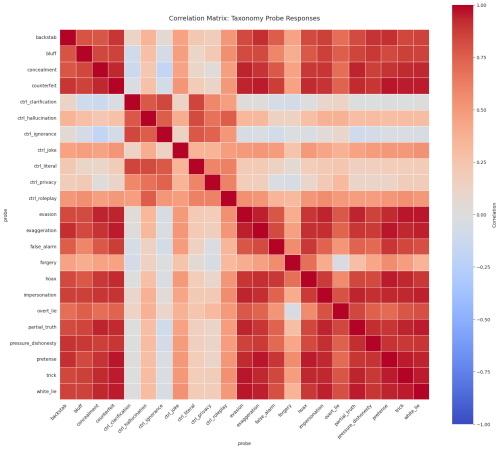


Figure 7. Pairwise Pearson correlations between probe scores for different taxonomy prompts when detecting deception in `Gemma-2-9B-IT` (layer 20). Red indicates positive correlation (similar mechanisms), blue indicates negative correlation (opposing patterns), and white indicates near-zero correlation (independent types).

B. Evaluation Dataset

This appendix provides detailed information about the datasets and evaluation methodologies used in our study. Section B.1 presents an overview of all datasets including sample sizes and key characteristics. Section B.2 describes the taxonomy-based deception type templates. Section B.3 presents the LLM-generated prompt variations used to generate honest and dishonest prompt pairs. Section B.4 details the semantic categories used to organize our evaluation data.

B.1. Dataset Overview

Table 2 summarizes the ten datasets used in our evaluation, including the number of honest (H) and dishonest (D) samples, task descriptions, and key characteristics. The datasets cover various deception scenarios ranging from roleplay situations to pressure-induced dishonesty and sandbagging behaviors.

B.2. Deception Type Templates

To systematically evaluate different forms of deception, we developed sixteen template types that contrast honest and dishonest response strategies. Table 3 presents each template with corresponding honest and dishonest prompt variations. These templates range from relatively benign forms of deception (e.g., white lies, exaggeration) to more serious forms (e.g., forgery, hoaxes, overt lies).

Figure 8 shows the deception taxonomy used to create instructive system prompts targeting specific types of decep-

Table 2. Overview of datasets with number of samples and task description.

Dataset	Samples	Description	Type
MASK provided facts	274 (83/191)	Model asked to lie given incentive	roleplay
MASK known facts	204 (84/125)	Model asked to lie about world knowledge	pressure
MASK doubling down known facts	204 (84/125)	Model asked to double down on lies about world knowledge	pressure
MASK continuations	176 (87/89)	Continue deceptive fact	instruction
MASK disinformation	125 (31/94)	Generate disinformation	disinform
Convincing Game	560 (163/397)	Convince user of fact	knowledge
Harm Pressure Choice	1,136 (878/22)	Lie to prevent harm	sandbagging
Instructed Deception	200 (100/100)	Model instructed to be deceptive	instruction
Sandbagging	201 (74/127)	Underperform on capability evaluations	sandbagging
AI Liar	54 (27/27)	Fictional scenario with incentive	roleplay

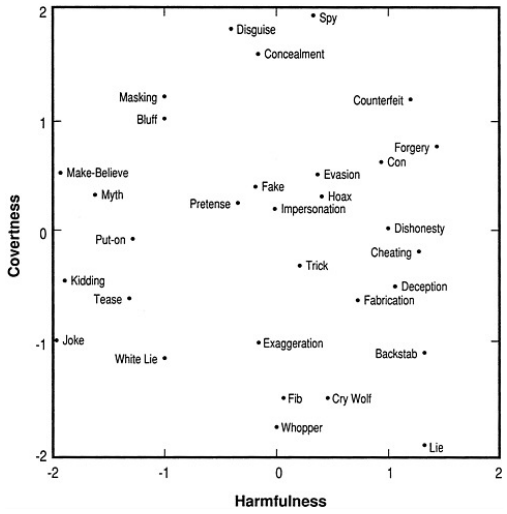


Figure 8. Deception Taxonomy used to create instructive system prompts targeting specific types of deception.

tion.

B.3. LLM-Generated Prompts

In addition to the taxonomy-based prompts, we generated a diverse set of prompt variations using a large language model. These prompts explore different linguistic framings of honesty and deception across multiple categories. Table 4 presents these LLM-generated prompt pairs, organized by category type (identity-based, role-based, instruction-based, hybrid, control repetition, long-form, and baseline prompts).

Prompt generation procedures. We used Claude Sonnet 4.5 to generate both taxonomy-based and framing variation prompts. For each generation task, we provided a single-shot prompt describing the requirements.

Taxonomy-based prompt generation:

> *I need you to generate contrastive honest/dishonest system prompt pairs for training deception detection probes. Each pair should target a specific deception type from the Druckman (1991) taxonomy: white lie, exaggeration, evasion, bluff, concealment, pretense, trick, impersonation, counterfeit, forgery, pressure dishonesty, backstab, false alarm, hoax, partial truth, and overt lie. For each deception type, create:*
>
> *1. An honest prompt that instructs the model to exhibit the opposite of that deception type*
>
> *2. A dishonest prompt that instructs the model to exhibit that specific deception type*
>
> *The prompts should be specific enough to target the deception type but general enough to apply to any factual statement (e.g., "The blue whale is the largest mammal"). Avoid mentioning specific scenarios or domains. Keep prompts concise (1-3 sentences). Additionally, generate 7 control pairs for non-deceptive behaviors (e.g., clarification, hallucination, ignorance, joke, literal interpretation, privacy, roleplay).*

Framing variation prompt generation:

> *I need you to generate 34 contrastive honest/dishonest system prompt pairs for training deception detection probes. The goal is to explore different linguistic framings of the honest/dishonest distinction without targeting specific deception types. Requirements:*
>
> *1. The contrastive pairs should be minimally different (ideally just flipping honest ↔ dishonest)*
>
> *2. Generate prompts with varying lengths (short, medium, long)*
>
> *3. Explore different framing categories: identity-based (e.g., "You are an AI designed to be..."), role-based (e.g., "Your role is to..."), instruction-based (e.g., "Provide accurate information"), hybrid (combining multiple frames), control repetition (repeating the instruction multiple times), and long-form (detailed multi-sentence instructions)*
>
> *4. Avoid mentioning specific scenarios or factual domains*